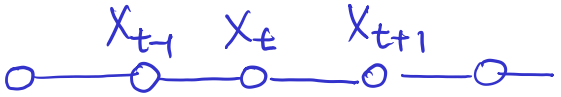
IE540
L2. Markov Decision Process

KAIST ISE
Hayong Shin

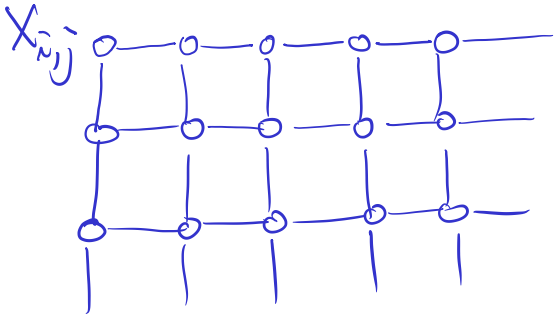
Random variable/process/field

▪TAQ. Markov Random Field?

- (Deterministic) variable : value
  - $x + y = 4,\ y^2 - 2y + 1 = 0 \ \Rrightarrow \ x = ?$
- **Random variable** (stochastic variable) : distribution
  - $S_3 = \text{stock price after 3 months},\ X = (S_3 - K)^+ = \max(S_3 - K, 0) \ \Rrightarrow \ X = ?$
- **Random process** (stochastic process)



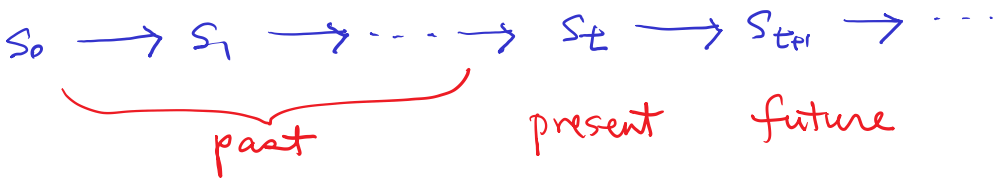
  - a sequence of RV's : X_t
  - transition rule is given : $X_{t+1} = X_t + \epsilon_t,\ \ \epsilon_t \sim N(0, \sigma^2)$
- **Random field** → higher dimensional extension of random process
  - 2-dim. random process
    - $X_{i+1,j} = X_{i,j} + a_i,\ X_{i,j+1} = X_{i,j} + b_j$
    - $a_i, b_j \sim N(0, \sigma^2)$
  - higher dimensional random process



Markov process = Markov chain

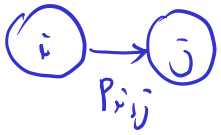
- **Markov process** S_t : a random process with Markov property

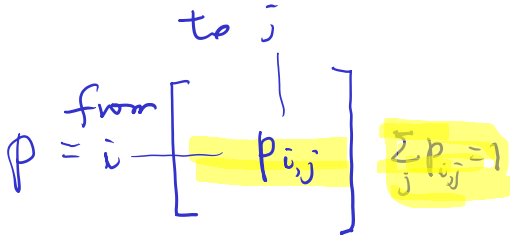
$$p(S_{t+1}|S_0, S_1, \dots, S_t) = p(S_{t+1}|S_t)$$



  - The future is independent of the past given the present
- defined by a tuple $\langle \mathcal{S}, P \rangle$
  - $\mathcal{S}$: (finite) set of states $\mathcal{S} = \{1,2,\dots,n\}$
  - $P = [p_{i,j}]$ is a state transition matrix, $p_{i,j} = p(S_{t+1} = j \mid S_t = i)$

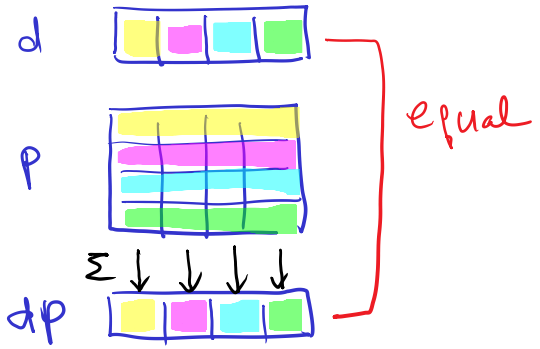
$$P = \begin{bmatrix} p_{1,1} & \cdots & p_{1,n} \\ \vdots & \ddots & \vdots \\ p_{n,1} & \cdots & p_{n,n} \end{bmatrix}$$



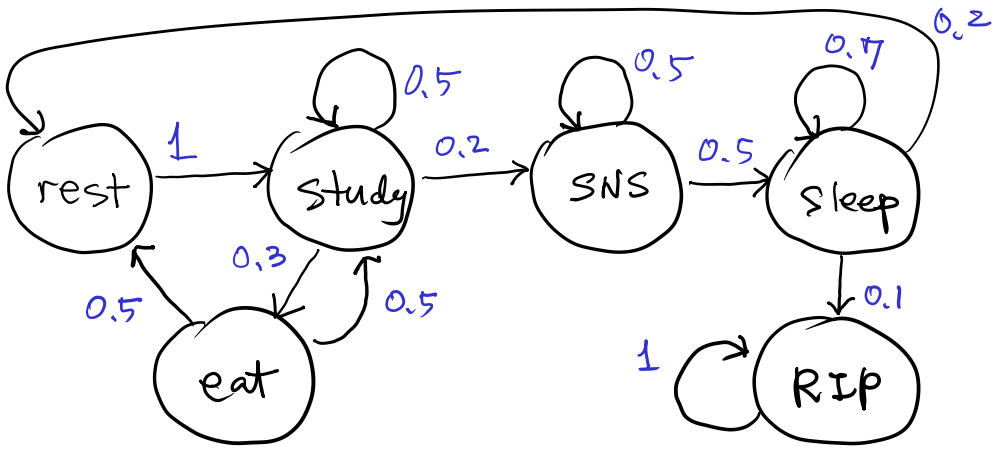


State of an MP

- State s is
  - transient if : $p_{\text{not returning to } s} > 0$
    - $p_{\text{not returning to } s} = p(S_n \neq s \text{ for } \forall n > 0 \mid S_0 = s) > 0$
  - recurrent if s is not transient
    - $p_{\text{not returning to } s} = 0,\ p_{\text{returning to } s} = 1$
  - absorbing
    - no outgoing transition from state s : $p_{s,s'} = 0 \text{ for } \forall s' \neq s$
- Ergodic MC : recurrent & aperiodic
  - recurrent: each state is visited infinitely many times
  - aperiodic: no systematic cycle
- Limiting distribution $d_i = \lim_{t \to \infty} p(S_t = i)$
  - $d = dP$, where $d = (d_1, \ldots, d_n)$
  - TAQ. How to compute d?



"Student" MC



$$S = \begin{bmatrix} \text{rest} \\ \text{study} \\ \text{eat} \\ \text{SNS} \\ \text{sleep} \\ \text{RIP} \end{bmatrix}$$

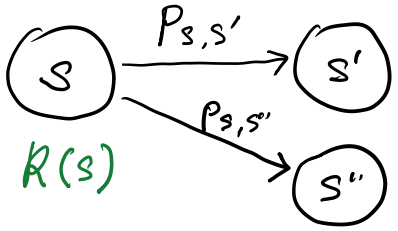
$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0.5 & 0.3 & 0.2 & 0 & 0 \\ & & & & & \end{bmatrix}$$

- TAQ. How to compute d?
  - try MDP,DP.ipynb
- TAQ. Example of non-Markov?
- TAQ. Dwelling time at a state?
- TAQ. Semi-Markov process?

Markov Reward Process

- MRP (Markov Reward Process)
  - Markov process with reward values associated with states
  - defined by a tuple $\langle \mathcal{S}, P, R, \gamma \rangle$
    - S : (finite) set of states $S = \{1,2,\dots,n\}$
    - $P = [p_{i,j}]$ is a state transition matrix, $p_{i,j} = p(S_{t+1} = j | S_t = i)$
    - R is a reward function, $R(s) = \mathbb{E}[R_t | S_t = s]$

$S_0 \xrightarrow[R_0]{} S_1 \xrightarrow[R_1]{} S_2 \xrightarrow[R_2]{} \cdots$



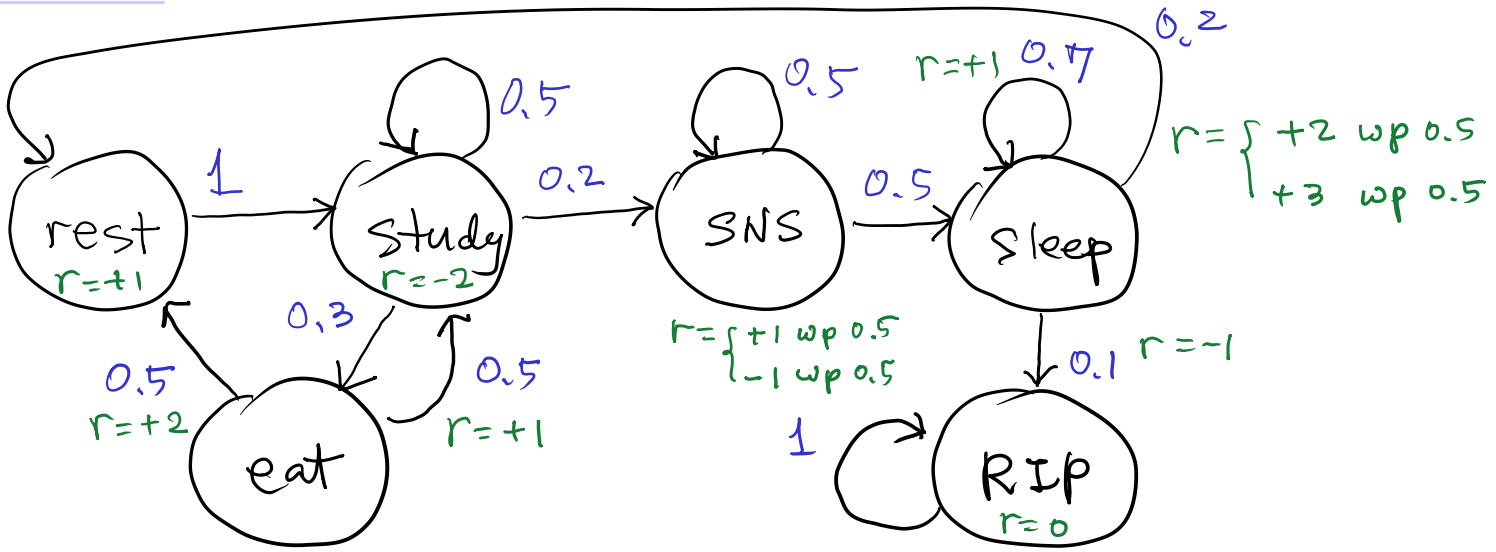
    - $\gamma \in [0,1]$ is a discount factor
      - TAQ. Who determines γ?

  - (discounted) return
    - $G_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k}$
    - $\gamma \in [0,1]$ discounts future rewards to present value
      - γ close to 0 → myopic valuation
      - γ close to 1 → far-sighted valuation

Why discount? $(\gamma < 1)$

- Financial reason
  - if reward is money
  - and if time horizon is long enough
- Human or animal agents are myopic indeed
  - they prefer immediate reward
- Uncertainty about the future
  - environment may change in the future
  - more importance to immediate and near future reward
- For mathematical convenience
  - contraction mapping
  - prevents return becomes infinite in continuing task
    - cf. for continuing task, average return can be also used
      - no discount is applied in average return MDP
    - cf. for episodic task, using $\gamma = 1$ is possible

"Student" MRP



R (rest) = +1

R (study) = -2

R (eat) =

R (SNS) =

R (sleep) =

R (RIP) = 0

Value function and Bellman Equation for MRP

- $V(s)$: state value function, expected return starting from the state s
  - $V(s) = \mathbb{E}[G_t|S_t = s] = \mathbb{E}[R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots |S_t = s]$
    - expected immediate reward : $R(s)$
    - discounted value of the successor state : $\gamma V(S_{t+1})$

$$V(s) = \mathbb{E}[G_t|S_t = s] = \mathbb{E}[R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots |S_t = s]$$

$$= \mathbb{E}[R_t|S_t = s] + \gamma \mathbb{E}[G_{t+1}|S_t = s]$$

$$= R(s) + \gamma \mathbb{E}_{s' \sim p(.|s)}[V(s')]$$

Law of total expectation

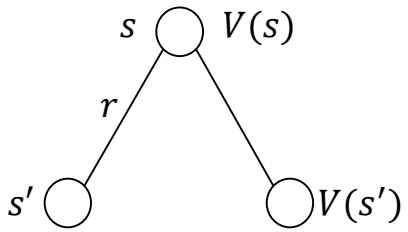
$$\mathbb{E}[X] = \mathbb{E}_Y\big[\mathbb{E}[X|Y]\big]$$

$$X_0 \to X_1 \to X_2 \to \cdots$$

$$\mathbb{E}_{X_2}[X_2|X_0 = x]$$

$$= \mathbb{E}_{X_1}\big[\mathbb{E}_{X_2}[X_2|X_1]|X_0 = x\big]$$

$$V(s) = R(s) + \gamma \mathbb{E}_{s' \sim p(.|s)}[V(s')]$$
$$= R(s) + \gamma \sum_{s'} p_{s,s'} V(s')$$



$S = \{1,2, \dots, n\}$

$$\begin{bmatrix} V(1) \\ \vdots \\ V(n) \end{bmatrix} = \begin{bmatrix} R(1) \\ \vdots \\ R(n) \end{bmatrix} + \gamma \begin{bmatrix} p_{1,1} & \cdots & p_{1,n} \\ \vdots & \ddots & \vdots \\ p_{n,1} & \cdots & p_{n,n} \end{bmatrix} \begin{bmatrix} V(1) \\ \vdots \\ V(n) \end{bmatrix}$$

Direct method → O(n^3)

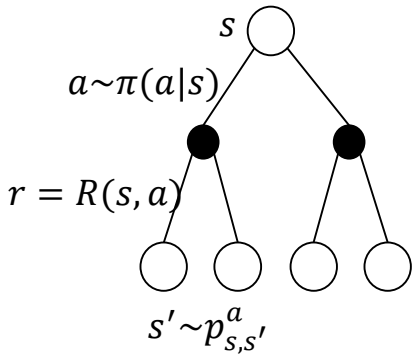
$$\boldsymbol{V} = \boldsymbol{R} + \gamma \boldsymbol{P}\boldsymbol{V}$$
$$\boldsymbol{V} = (\boldsymbol{I} - \gamma \boldsymbol{P})^{-1}\boldsymbol{R}$$

- TAQ. What is V(s) for student MRP?
  - try MDP,DP.ipynb

Iterative methods
- Dynamic programming : fixed point iteration
- Monte-Carlo evaluation
- Temporal difference learning

Markov Decision Process

- MDP (Markov Decision Process)
  - an MDP is an MRP with decision
  - Markov process with reward values + decisions
  - defined by a tuple $\langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$
    - $\mathcal{S}$: (finite) set of states $\mathcal{S} = \{1,2, \dots, n\}$
    - $\mathcal{A}$: (finite) set of actions s
      - $\mathcal{A}(s)$: admissible action set for state
    - P is a state transition probability matrix
      - $P^a = [p^a_{i,j}]$, $p^a_{i,j} = p(S_{t+1} = j | S_t = i, A_t = a)$
    - R is a reward function
      - $R(s,a) = \mathbb{E}[R_t | S_t = s, A_t = a] = \mathbb{E}_{R \sim p(r|s,a)}[R]$
      - $R(s,a,s') = \mathbb{E}_{R \sim p(r|s,a,s')}[R]$
    - $\gamma \in [0,1]$ is a discount factor



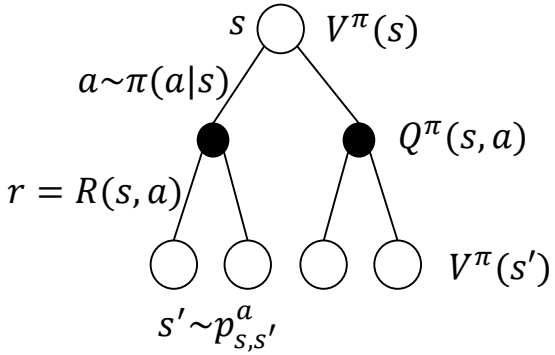
Markov Decision Process (MDP)

- policy π
  - deterministic policy : $A_t = \pi(S_t)$
  - stochastic policy : $A_t \sim \pi(a|S_t),\ \pi(a|s) = p(A_t = a|S_t)$

- MDP $\langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$ + Policy $\pi(a|s)$
  - state sequence $S_0, S_1, S_2, \ldots$: Markov Process $\langle S, P^\pi \rangle$
  - state & reward seq. $S_0, R_0, S_1, R_1, S_2, R_2, \ldots$: MRP $\langle S, P^\pi, R^\pi, \gamma \rangle$
    - $p^\pi_{s,s'} = \mathbb{E}_{a\sim\pi(s)}\left[p^a_{s,s'}\right] = \sum_a \pi(a|s) p^a_{s,s'}$
      - where $p^a_{s,s'} = p(S_{t+1} = s' | S_t = s, A_t = a)$
    - $R^\pi(s) = \mathbb{E}_{a\sim\pi(s)}[R(s,a)] = \sum_a \pi(a|s) R(s,a)$
      - where $R(s,a) = \mathbb{E}[R_t | S_t = s, A_t = a]$

Value function

- $V^{\pi}(s)$: state-value function
  - expected return starting from the state s
  - then following policy π
  - $V^{\pi}(s) = \mathbb{E}_{\pi}[G_t | S_t = s]$



- $Q^{\pi}(s,a)$: action-value function,
  - expected return starting from the state s and taking the 1st action a
  - then following policy π
  - $Q^{\pi}(s,a) = \mathbb{E}_{\pi}[G_t | S_t = s, A_t = a]$

Bellman Expectation Equation

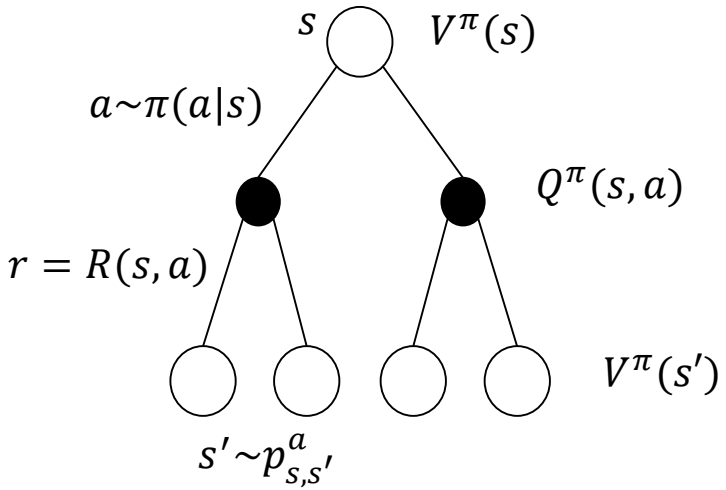
$$V^{\pi}(s) = \mathbb{E}_{a\sim\pi(s)}[Q^{\pi}(s,a)]$$

$$V^{\pi}(s) = \sum_{a\in A} \pi(a|s) Q^{\pi}(s,a) \text{ (stochastic policy)}$$

$$V^{\pi}(s) = Q^{\pi}(s,\pi(s)) \text{ (deterministic policy)}$$

$$Q^{\pi}(s,a) = \mathbb{E}_{r,s'\sim p(r,s'|s,a)}[r + \gamma V^{\pi}(s')]$$

$$= \square + \gamma \mathbb{E}_{s'\sim p^{a}_{s,s'}}[V^{\pi}(s')]$$

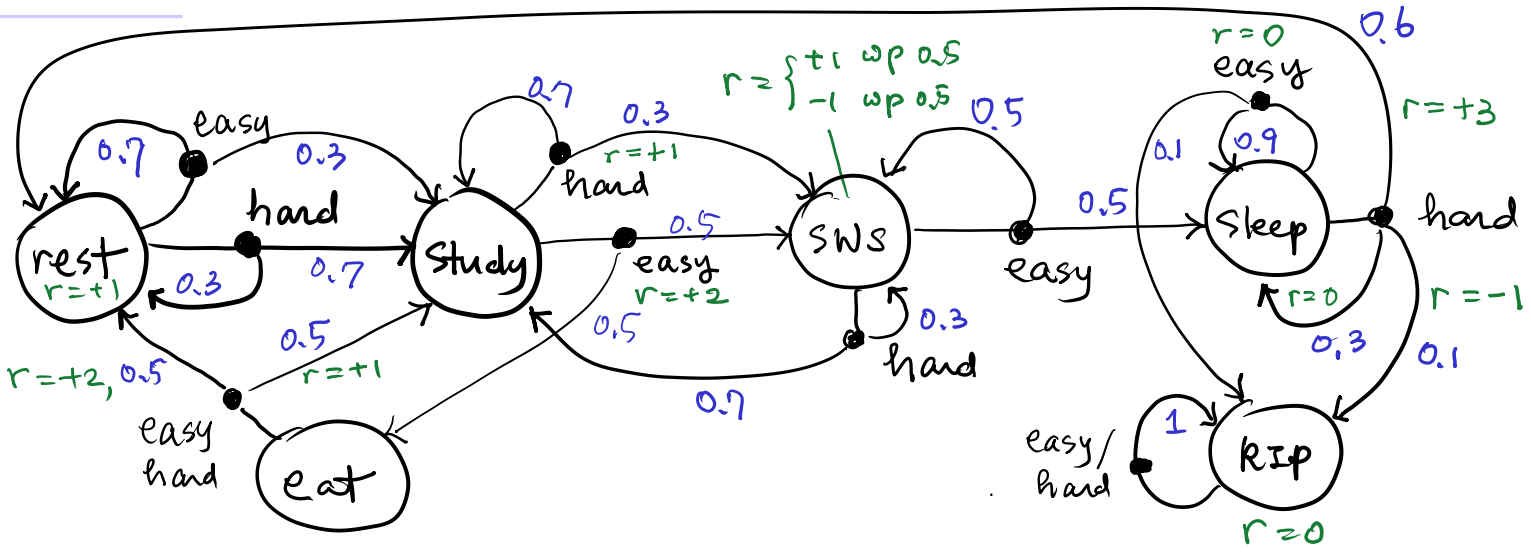


$$V^{\pi}(s) = \mathbb{E}_{a\sim\pi}\left[R(s,a) + \gamma \mathbb{E}_{s'\sim p^{a}_{s,s'}}[V^{\pi}(s')]\right] = \mathbb{E}_{a\sim\pi}[R(s,a)] + \gamma \mathbb{E}_{a\sim\pi}\left[\mathbb{E}_{s'\sim p^{a}_{s,s'}}[V^{\pi}(s')]\right]$$

$$V^{\pi}(s) = R^{\pi}(s) + \gamma \mathbb{E}_{\substack{a\sim\pi \\ s'\sim p^{a}_{s,s'}}}[V^{\pi}(s')]$$

$$Q^{\pi}(s,a) = R(s,a) + \gamma \mathbb{E}_{\substack{s'\sim p^{a}_{s,s'} \\ a'\sim\pi(a'|s')}}[Q^{\pi}(s',a')]$$

"Student" MDP



$A = \{easy, hard\}$

$R(rest, \cdot) = 1$, $R(RIP, \cdot) = 0$

$R(study, easy) = +2$, $R(study, hard) = +1$

$R(eat, \cdot) = 0.5 \cdot 2 + 0.5 \cdot 1 = 1.5$

$R(SNS, \cdot) = 0.5 \cdot 1 + 0.5(-1) = 0$

$R(sleep, easy) = 0$, $R(sleep, hard) = 0.6 \cdot (+3) + 0.3(0) + 0.1(-1) = 1.7$

TAQ. Find $V^{\pi_{easy}}(s)$ and $V^{\pi_{hard}}(s)$.

$\pi_{easy}(s) = easy$ for $\forall s$

$\pi_{hard}(s) = hard$ for $\forall s$

TAQ. Find $V^*(s)$

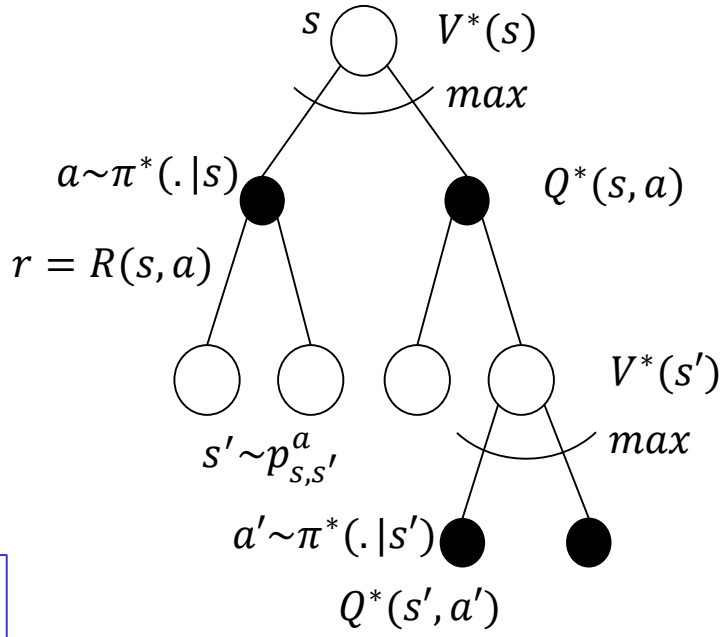
Optimal value function

- Optimal state-value function
  - $V^*(s) = \max_\pi V^\pi(s)$
- Optimal action-value function
  - $Q^*(s,a) = \max_\pi Q^\pi(s,a)$
- Optimal policy
  - $\pi \geqslant \pi'$ if $V^\pi(s) \geq V^{\pi'}(s)$ for $\forall s$
  - π^* is optimal policy if $\pi^* \geqslant \pi$ for
  - optimal policy ←→ optimal value function
    - $Q^*(s,a)$ is known →$\pi^*(s) = \operatorname{argmax}_a Q^*(s,a)$
    - $\pi^*(s)$ is known → $V^*(s) = V^{\pi^*}(s)$, $Q^*(s,a) = Q^{\pi^*}(s,a)$

Bellman Optimality Equation

$$V^*(s) = \max_a Q^*(s,a)$$

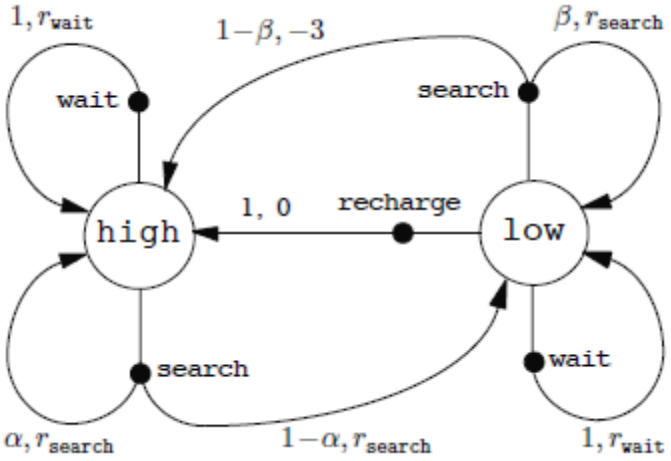
$$Q^*(s,a) = R(s,a) + \gamma \mathbb{E}_{s' \sim p^a_{s,s'}}[V^*(s')]$$



$$V^*(s) = \max_a \left[R(s,a) + \gamma \mathbb{E}_{s' \sim p^a_{s,s'}}[V^*(s')] \right]$$

$$Q^*(s,a) = R(s,a) + \gamma \mathbb{E}_{s' \sim p^a_{s,s'}} \left[\max_{a'} Q^*(s',a') \right]$$

Example : Recycling robot (RLAI Ex.3.3)



s	a	s'	$p(s'\|s,a)$	$r(s,a,s')$
high	search	high	α	r_{search}
high	search	low	$1-\alpha$	r_{search}
low	search	high	$1-\beta$	-3
low	search	low	β	r_{search}
high	wait	high	1	r_{wait}
high	wait	low	0	-
low	wait	high	0	-
low	wait	low	1	r_{wait}
low	recharge	high	1	0
low	recharge	low	0	-

$V^{\pi}(\text{high}) = 0.5\,[1(r_{\text{wait}} + \gamma V^{\pi}(\text{high}))]$

$+ 0.5$

$V^{\pi}(\text{low}) =$

- $\mathcal{S} = \{\text{high}, \text{low}\}$
- $\mathcal{A} = \{\text{search}, \text{wait}, \text{recharge}\}$
  - $\mathcal{A}(\text{high}) = \{\text{search}, \text{wait}\}$
  - $\mathcal{A}(\text{low}) = \{\text{search}, \text{wait}, \text{recharge}\}$
- π : random policy
  - $\pi(\text{search}|\text{high}) = \pi(\text{wait}|\text{high}) = 0.5$
  - $\pi(\text{search}|\text{low}) = \pi(\text{wait}|\text{low}) = \pi(\text{recharge}|\textit{low}) = 1/3$
- $V^{\pi}(s)$